

Chapter 23 — Assessment of Urologic and Surgical Outcomes

- **Outcomes research** has grown over 25 years but is an **umbrella term without a consistent definition**; one view: the effectiveness of public-health interventions and health services on patient outcomes.
- **Agency for Healthcare Research and Quality (AHRQ)**: research into the **end results** of healthcare practices — effects patients experience and care about (e.g., ability to function); in chronic disease, where cure is not always possible, **quality of life as well as mortality**. Linking care received to outcomes makes it the key to monitoring and improving **quality of care**.
- **Urological end points** are many — quality of life, clinical effectiveness, cost, quality of care, preferences, appropriateness, access, health status — hence the difficulty of definition.
- **Health services research (HSR)**, often used interchangeably: multidisciplinary study of how social factors, financing, organisational structures and processes, technologies and personal behaviour affect access, quality, cost and ultimately health. It misses the clinical side → urological outcomes research = **HSR + clinical epidemiology + comparative effectiveness research + (partly) clinical trials**.

ESTABLISHING A CONCEPTUAL FRAMEWORK FOR ASSESSING THE EFFECTIVENESS OF TREATMENT AND IMPROVING CARE IN UROLOGY

- **Institute of Medicine (IOM) — quality of care**: the degree to which health services increase the likelihood of desired outcomes and are **consistent with current professional knowledge**. Affected by access, clinical effectiveness, integration (coordination, continuity), cultural competence and comprehensiveness → need for a conceptual model.
- **Donabedian model** — the most accepted framework: quality measured in three dimensions, **structure, process and outcome**; elements of each can and should be measured.

DONABEDIAN MODEL — STRUCTURE, PROCESS, OUTCOME

	Structure	Process	Outcome
Definition	Context in which care is delivered	Activities by professionals and systems to deliver care	Effect of care on patients or populations
Examples	Procedure volume, subspecialty training, nurse-to-bed ratio, "closed" intensive care unit, technology	Appropriate imaging and laboratory use; antibiotic and deep vein thrombosis (DVT) prophylaxis; technical operative steps	Mortality, length of stay, readmission, satisfaction, quality of life, cost-effectiveness, utilisation
Strengths	Easy, cheap (administrative, public data); undoubtedly influences quality	Proven link to outcome where evidence robust; detects problems before outcome differences appear ; cheaper; less affected by case mix if well specified; Donabedian thought it the most reflective measure	Most meaningful to patients and clinicians; some argue it should be the sole focus
Weaknesses	Misses care actually delivered by the provider; largely non-modifiable	Valid only where evidence links process to outcome	Confounded by patient factors (age, comorbidity) → needs risk adjustment

- **Guidelines generate process measures**: AUA (American Urological Association) and EAU (European Association of Urology) guidelines — e.g., **24–36 months of androgen deprivation therapy (ADT) with**

external beam radiotherapy for high-risk localised prostate cancer (CaP) — used by Medicare's physician quality reporting system and private payers.

- **Risk-adjustment trap:** without case-mix adjustment, providers treating high-risk patients look worse → an **economic disincentive to treat high-risk patients**.

Concept — *Structure enables process and process produces outcome; the further along that chain a measure sits, the more it matters to patients and the more it is contaminated by case mix.*

LONG-TERM DISEASE OUTCOMES THAT ARE COMMONLY ASSESSED IN UROLOGY

- Patients and urologists focus on **outcomes**. Surgeons favour **morbidity and mortality** — the "hardest", theoretically easiest end points; patients and other stakeholders add "**softer**" end points: patient-reported outcomes (PROs — symptoms, bother), economics, satisfaction.

Overall Mortality

- **The most objective end point** — alive or dead is rarely disputed; key in population-based epidemiology.

SOURCES OF VITAL STATUS (UNITED STATES)

Source	Features
National Death Index (NDI)	National Center for Health Statistics within the Centers for Disease Control and Prevention; from each state's vital statistics; updated annually, ~1-year lag . Researcher submits identifiers (name, Social Security number, date of birth, race, sex, marital status, states of residence and birth) → date and place of death, certificate number, cause of death . Highly accurate; deaths after 1979 only
Social Security Administration (SSA)	For deaths before 1979 ; less accurate than the NDI
Credit agencies, internet databases	May reach internationally; less comprehensive

- **Expression:** count, ratio, proportion or rate; **proportion or rate** suits clinical studies. A proportion ignores time → **mortality rate** (proportion dying over a specified period) preferred. Example: in a trial of second- or third-line therapy for metastatic castration-resistant CaP, all may die during the study → compare **1-, 2- and 5-year rates**.
- **Time-to-event (survival) analysis** — usable for any binary end point; handles **varying follow-up and loss to follow-up** through **censoring**: a participant known to be event-free up to last follow-up, study end, or a competing event that makes the outcome impossible (e.g., fatal unrelated myocardial infarction before recurrence) "survives" to that point and is then censored.
- **Kaplan–Meier curves** display survival estimates at time points incorporating both events and censoring.
- **Objective ≠ unbiased:** design introduces lead-time and length-time bias.

BIASES THAT DISTORT SURVIVAL AND PROGRESSION END POINTS

Bias	Mechanism	Urological example
Lead-time	Screening or new diagnostics advance the date of diagnosis ; lead time = detection to clinical presentation; survival appears longer though death is not delayed	Kidney cancer: incidence doubled with abdominal CT (computed tomography), 5-year survival rose, mortality rate unchanged ; claimed survival gains from prostate-specific antigen (PSA) screening
Length-time	Each screen falls at a random point; slow-growing tumours have longer preclinical	Indolent low-grade CaP over-represented among screen-detected cancers

Bias	Mechanism	Urological example
	phases and are over-detected; they are less lethal anyway	
Stage migration (Will Rogers phenomenon)	A more sensitive test reclassifies patients to a higher stage without changing outcome → apparent improvement	Prostate-specific membrane antigen-positron emission tomography (PSMA-PET) — see below
Attribution	Cause of death misassigned	Disease-specific mortality (below)
Detection	One group more likely to have an event detected (more imaging or follow-up)	Progression in observational or retrospective studies; less likely in protocol-driven trials

Figure legends.

- **Fig. 23.1 Lead-time bias:** cancer onset at 50. Unscreened — clinical diagnosis at 70, death at 75 → survival 5 years. Ineffective screening — detected at 60, death still at 75 → "survival" 15 years, all 10 extra years lead time. Effective screening — detected at 60, death at 80 → 20 years, of which only 5 are real gain.
- **Fig. 23.2 Length-time bias:** each patient's arrow runs from onset to clinical presentation; slow tumours have longer **detectable preclinical periods (DPCP)** and longer survival after diagnosis, so a screen catches more of them. Without screening, mean DPCP 6 and survival 4 years; with screening, 8 and 6 years — apparent prolongation.

Disease-Specific Mortality

- Overall mortality is the hardest but **crudest** end point: it ignores intercurrent illness and is **not germane in benign or low-mortality disease** → disease-specific mortality.
- **Definition:** deaths attributed directly to the disease — **harder to define than it appears.**
- **Perioperative deaths** are instinctively not attributed to the disease, yet arguably are disease- and treatment-related. In an early-PSA-era Surveillance, Epidemiology, and End Results (SEER) analysis, **most deaths within a month of CaP surgery were attributed to other causes**; reattribution would noticeably raise disease-specific mortality in a low-mortality cancer.
- **Medical treatment:** ADT raises cardiovascular risk and death — is a cardiac death on ADT a CaP death? Pneumonia and sepsis in an immobile metastatic patient — cancer-related or not?
- **Attribution bias — the greatest limitation:** a documented cancer is often written on the death certificate even when the death is unrelated.
 - **CaP:** death certificates agree with independent chart review in most cases, but **disagree in about one in ten**; similar across several US regions.
 - **Urinary tract tumours (SEER):** a substantial share of deaths attributed to non-urological causes; **more advanced stage → more often attributed to cancer; site matters** — renal pelvic tumours far less often recorded as cancer deaths than similar-stage kidney tumours.
 - Studies examine **in-hospital deaths** only; deaths at home or in nursing facilities are harder still.
- **Bottom line:** usually reliable, but attribution bias and misclassification can alter conclusions.
- **PSMA-PET:** high sensitivity for recurrent or oligometastatic disease even at low PSA, especially after biochemical recurrence → changes management in a majority (ADT or systemic therapy, salvage prostate-bed radiotherapy, metastasis-directed ablation). Earlier detection than CT and bone scintigraphy makes **progression-free survival (PFS) appear longer** because (1) smaller-volume disease has a different natural history and better treatment response, and (2) earlier detection with **reclassification to a higher stage without changing outcome — stage migration (Will Rogers phenomenon)** → PFS improves, **overall survival does not.**

Other Binary "Survival" Outcomes

- **Any binary outcome** can become a survival end point analysed by Kaplan–Meier and proportional hazards: **metastasis-free, radiological progression-free, symptom-free, biochemical-free survival**.
- Used in **benign disease** and in CaP, where deaths are rare or late; relevant outcomes differ by disease.
- **Objectivity varies**: urodynamics often fails to reproduce stress or urgency incontinence, and no agreed leakage threshold defines incontinence; patient-reported alternatives (pad use, symptom scores) are less "objective" than radiological or serum measures.

Proxy End Points

- Mortality takes years and is almost irrelevant in benign conditions (incontinence, stones) → **clinically relevant or proxy end points**.
- **Prentice criteria (1989) for a valid surrogate**: (1) treatment associated with the true end point (overall or disease-specific survival); (2) treatment associated with the surrogate; (3) surrogate associated with the true end point; (4) **the surrogate explains the full effect of treatment on the true end point**.
- **Few urological proxies meet all four**.

Disease Progression/Recurrence

- **PFS** is a common oncological proxy; research requires **standardised radiological definitions** of change in tumour burden.
- **Evolution**: World Health Organization criteria (1981) → study-by-study modifications, confusion and conflicting results → European Organisation for Research and Treatment of Cancer (EORTC), US National Cancer Institute (NCI) and National Cancer Institute of Canada working group → **Response Evaluation Criteria In Solid Tumours (RECIST), 2000**: minimum measurable size, up to ten lesions (five per organ), **unidimensional** measurement.
- **Limitations of RECIST**: (1) response only, poor measure of progression; (2) no magnetic resonance imaging (MRI) or PET; (3) no lymph nodes; (4) cannot assess targeted non-cytotoxic drugs → **RECIST 1.1 (2009)**.
- **RECIST 1.1 measurable lesion**: **≥10 mm on CT, 20 mm on chest radiograph**, 10 mm clinically with callipers; lymph node **≥15 mm short axis** on CT. **Not ultrasound; not tumour markers alone** — except **PSA response** (Prostate Cancer Clinical Trials Working Group), usable alongside RECIST.
- **Target lesions**: one to five, the **largest**, representative of involved organs and reproducibly measurable; their **sum of diameters** is followed from baseline → radiological PFS, duration of response, overall response rate.

TABLE 23.1 RECIST CRITERIA

Category	Target lesions	Non-target lesions
Complete response	All disappear; pathological nodes <10 mm short axis	All disappear, tumour markers normal ; nodes <10 mm
Partial response	≥30% decrease in sum, vs baseline	—
Stable disease	Neither partial response nor progressive disease, vs smallest sum on study	—
Non-complete response / non-progressive disease	—	≥1 non-target persists and/or marker above normal
Progressive disease	≥20% increase vs smallest sum on study (nadir) and ≥5 mm absolute ; or any new lesion	Unequivocal progression and/or new lesions

From Eisenhauer EA, Therasse P, Bogaerts J, et al. Revised RECIST guideline (version 1.1). Eur J Cancer 2009.

- **What it tests:** the reference points differ — **response is measured against baseline, progression against nadir**; progression needs both the relative and the absolute increase.
- RECIST standardises progression but **does not remove detection bias** (see bias table).
- **Observer variability:** even for stone burden on CT, radiologists disagreed with one another and with their own earlier readings to a substantial degree → radiological outcomes are **less objective than assumed**; hence **central review** of imaging or pathology in prospective trials.

Intermediate Clinical End Points in Cancer of the Prostate

- CaP's long natural history → overall-survival trials can exceed a decade, and many relapsing men never die of CaP → a valid **intermediate clinical end point (ICE)** accelerates drug development and saves resources.
- **Intermediate Clinical End Points in Cancer of the Prostate (ICECaP) working group** tested PSA-based ICEs (time to biochemical failure, PSA doubling time, PSA nadir, end-of-treatment PSA) and metastasis-free survival (MFS).

PSA-based ICEs	Metastasis-free survival
Did not track overall survival — treatment intensification worsened other-cause mortality without commensurate CaP-specific benefit	Strongly correlated with overall survival across trials → proposed as the valid ICE for localised CaP

Receipt of Secondary Therapy

- Reported in malignant and benign conditions; **looks objective but is biased**.
- **Adjuvant or secondary?** Radiotherapy or hormonal therapy after surgery for high-risk disease is ambiguous.
- **Subjective initiation:** men with biochemical recurrence after radical prostatectomy often elect hormonal therapy or radiotherapy although the "recurrence" may not be clinically meaningful — **"discretionary" treatment**.

Concept — *The harder the end point, the slower and cruder it is; every proxy trades time for validity, the Prentice criteria test whether the trade was honest, and in prostate cancer only metastasis-free survival has passed.*

COMMONLY ASSESSED SHORT-TERM OUTCOMES

Assessing Surgical Complications

- Systems: **Common Terminology Criteria for Adverse Events (CTCAE)**, its **patient-reported version (PRO-CTCAE)**, and **Clavien–Dindo**.
- **Complication:** any deviation from the normal or expected postoperative course; degrees and unexpectedness vary → standardised systems for clinical and research use.

COMPLICATION-REPORTING SYSTEMS COMPARED

	CTCAE v5.0	PRO-CTCAE	Clavien–Dindo
Origin	Early 1980s; cancer treatment; NCI cooperative groups and industry	Patient-reported companion to CTCAE	Clavien 1992; Dindo 2004 modification
Reporter	Clinician	Patient	Clinician
Basis			

	CTCAE v5.0	PRO-CTCAE	Clavien–Dindo
	Severity per organ system, grades 1–5	Presence, frequency, severity, interference; library of 78 symptomatic adverse events	Therapy required to treat the complication; 5 grades, 7 strata
Strength	Simple; suits novel agents with unexpected serious toxicity	Longitudinal patient view; reliable, valid, good compliance, including urological trials	Extensively validated; lower subjectivity
Limitation	Unrefined, not surgery-specific , lacks granularity for comparative effectiveness	Scores do not map to CTCAE grades ; not a safety signal for dose decisions	Residual subjectivity (below)

Common Terminology Criteria for Adverse Events

TABLE 23.2 CTCAE SYSTEM FOR CLASSIFICATION OF SURGICAL AND MEDICAL PROCEDURES

Grade	Definition
1	Asymptomatic or mild; clinical or diagnostic observation only; no intervention
2	Moderate; minimal, local or non-invasive intervention; limits age-appropriate instrumental activities of daily living (ADLs)
3	Severe or medically significant, not immediately life-threatening ; hospitalisation or prolongation; disabling; limits self-care ADLs
4	Life-threatening ; urgent intervention
5	Death

- **Trap:** grades 2 and 3 are separated by the ADL affected — instrumental versus self-care.

Patient-Reported Outcomes Version of the Common Terminology Criteria for Adverse Events

- Clinician-reported adverse events and patient-reported quality of life **diverge**: patients under-report because of time pressure, not wishing to complain, and fear of losing eligibility, dose reduction or removal from protocol.
- **Use CTCAE grades for clinical decisions**; use PRO-CTCAE to describe **aggregate safety and tolerability**.

Clavien–Dindo System of Classifying Complications

- Grades by the **risk and invasiveness of the therapy** needed, minimising subjective judgement of severity. Dindo's modification (four to five grades, seven strata) adds **whether general anaesthesia was needed** and whether there was **organ failure or intensive care unit (ICU) admission**; validated, interobserver variability tested.

TABLE 23.3 CLAVIEN–DINDO CLASSIFICATION OF COMPLICATIONS

Grade	Definition	Example
I	Deviation without pharmacological, surgical, endoscopic or radiological treatment; antiemetics, antipyretics, analgesics, diuretics and similar allowed; wound infection opened at bedside	Ileus after cystectomy managed with observation and intravenous fluids (not parenteral nutrition)
II	Needs other pharmacological treatment	Bleeding after nephrectomy needing transfusion , no ICU
IIIa	Surgical, endoscopic or radiological intervention without general anaesthesia	Pelvic collection after prostatectomy drained percutaneously under local anaesthesia and mild sedation
IIIb	Intervention under general anaesthesia	Any unplanned return to theatre under general anaesthesia

Grade	Definition	Example
IVa	Life-threatening (including central nervous system*), intermediate care or ICU; single-organ dysfunction	Isolated myocardial infarction after PCNL (percutaneous nephrolithotomy) → cardiac ICU
IVb	As IVa with multiorgan dysfunction	Myocardial infarction and hypotension after partial nephrectomy → renal failure, acute respiratory distress syndrome, ventilation, pressors, dialysis
V	Death	Any postoperative death

*Brain haemorrhage, ischaemic stroke, subarachnoid bleeding; **excludes transient ischaemic attack**. Modified from Dindo D, Demartines N, Clavien PA. Ann Surg 2004.

- **Hinges:** I/II — drug beyond the permitted list; IIIa/IIIb — general anaesthesia; IVa/IVb — one organ or several.
- **Limitations:** (1) residual subjectivity — a recognised rectal injury at prostatectomy graded I (prolonged stay) or III (repair under general anaesthesia); (2) the same intervention under local or general anaesthesia by institution → variability in grades III and IV; (3) **two complications of equal grade** in one patient not captured; (4) same complication managed differently (inferior vena cava filter versus heparin for DVT).

Assessing Risk for Surgical Complications

- Beyond procedure-specific factors, three characteristics predict complications across all surgery: **functional status, comorbidity and frailty**.

Functional Status

- **Definition:** ability to perform normal daily activities to meet basic needs, fulfil roles and maintain health — effectively the ADLs.
- **Six essential ADLs:** eating, dressing, walking or transferring, bathing, toileting, continence.
- Matters in urological oncology trials for **inclusion criteria and outcomes**.
- **Karnofsky Performance Scale (KPS, 1949):** clinician-scored **100% (normal) to 0% (dead)**; acceptable inter- and intraobserver reliability.
- **Eastern Cooperative Oncology Group (ECOG) scale: 0 (fully active) to 5 (dead)**; performs as well as KPS.
- Future: **wearable technology** for real-time functional data.

TABLE 23.4 COMPARISON OF KARNOFSKY PERFORMANCE SCORE TO ECOG GRADE

ECOG	ECOG status	KPS	Karnofsky status
0	Fully active, no restriction	100	Normal, no complaints
1	Restricted in strenuous activity; ambulatory; light work	80–90	Normal activity (minor symptoms, or with effort)
2	Ambulatory, all self-care, no work; up >50% of waking hours	60–70	Self-care, cannot work → occasional assistance
3	Limited self-care; bed or chair >50% of waking hours	40–50	Considerable assistance → disabled, special care
4	Completely disabled; totally confined to bed or chair	10–30	Severely disabled → very sick → moribund
5	Dead	0	Dead

Modified from Oken MM, Creech RH, Tormey DC, et al. Am J Clin Oncol 1982.

- **Trap:** ECOG 2 versus 3 turns on **50% of waking hours** out of bed; KPS 70 (self-care, no work) = ECOG 2.

Comorbidity

- Term coined by **Feinstein (1970)**: any ailment coexisting with the studied disease.
- Comorbidity **influences treatment choice → confounder**; ignoring it causes **selection bias**.
- **Classic example — TURP (transurethral resection of the prostate)**: a large administrative-data study found higher long-term mortality after TURP than open prostatectomy without controlling for comorbidity; a chart-review study adjusting for age and comorbidity found **no difference in 5-year mortality**, later confirmed.
- **Charlson Comorbidity Index** — best known; weighted sum; developed to predict **1-year mortality in medical inpatients**, later **3–5-year mortality after general surgery**; short form and an administrative-data adaptation exist; other validated scales also used.

Charlson weight	Conditions
1	Myocardial infarction, congestive heart failure, peripheral vascular disease, cerebrovascular disease, dementia, chronic pulmonary disease, connective tissue disease, peptic ulcer, mild liver disease, diabetes
2	Hemiplegia, moderate–severe renal disease, diabetes with end-organ damage, any malignancy
3	Moderate–severe liver disease
6	Metastatic cancer, acquired immunodeficiency syndrome (AIDS)

Frailty

- **Definition:** clinically recognisable vulnerability from ageing-related decline in reserve across multiple systems → more falls, hospitalisation, disability, death.
- **Fried phenotype:** frail = **≥3 of 5** — low grip strength, low energy, slow walking, low physical activity, unintentional weight loss. Independently predicted falls, functional decline, hospitalisation and death over 3 years (Cardiovascular Health Study); later predicted **postoperative complications, length of stay and discharge to a skilled nursing facility**.
- Correlates with outcome after **cystectomy, prostatectomy and prolapse surgery** → control for it in non-randomised studies.
- **Criticism:** Fried's model is essentially **sarcopenia**; Charlson and the American Society of Anesthesiologists (ASA) classification have not reliably predicted postoperative morbidity and mortality.

FRAILITY INDICES

Index	Model	Size and use
Fried phenotype	Physical phenotype (sarcopenia)	5 criteria, ≥ 3 = frail
Canadian Study of Health and Aging Frailty Index (CSHA-FI)	Accumulating deficits incl. comorbidity and quality of life	~70 items — impractical
Modified Frailty Index (mFI)	Accumulating deficits	11 items ; validated in urological cancer surgery
Simplified Frailty Index (sFI)	Accumulating deficits	Shorter still; correlates with radical cystectomy outcomes

Pain

- **Three approaches:** (1) verbal or written description or rating — **used in most urology**; (2) rating its effect on observable behaviour, by patient or observer; (3) comparison with an experimentally induced stimulus — laboratory only. Research should use validated instruments.

Measure	Format	Note
Oral morphine equivalents	Analgesic consumption converted	Looks objective but biased by pain tolerance — some avoid analgesia despite pain
Verbal descriptor scale	"No pain" → "worst imaginable"	Adequate clinically; hard to use in research
Visual analogue scale (VAS)	10 cm line , anchored ends; mark measured in mm → 0–100	Simple, effective in research and clinic; variants: 0–10 numerical, descriptors along the line
McGill Pain Questionnaire; Brief Pain Inventory	20 items each	Detailed; suited to research
Patient-Reported Outcomes Measurement Information System (PROMIS) pain bank	National Institutes of Health (NIH) item bank	Validated, reliable

- **Multidisciplinary Approach to the Study of Chronic Pelvic Pain (MAPP) Research Network** (National Institute of Diabetes and Digestive and Kidney Diseases): studies **urological chronic pelvic pain syndromes** (chronic prostatitis, interstitial cystitis/bladder pain syndrome) across pain, interference, psychological stress and sexual function. Best two indices: **Genitourinary Pain Index** (pain) and **Interstitial Cystitis Symptom Index** (urinary symptoms).

Concept — *Complications are graded by the treatment they demand, not by how severe they looked, and risk is predicted by the reserve the patient brought — function, comorbidity, frailty — which must be adjusted for in every non-randomised comparison.*

PATIENT-REPORTED OUTCOMES

- Many interventions aim to relieve symptoms, but **symptom assessment differs from health-related quality of life (HRQoL)**.
- **Symptoms** are the physical manifestations of disease, assessed in ways that suit the condition.
- **Clinicians cannot measure patient-centred outcomes from the consultation:** patients minimise to please, physicians miss severity. Compared with a validated survey, physicians **consistently underestimated urinary, sexual and bowel dysfunction, fatigue and bone pain** → collect directly from patients.

Symptom scale	HRQoL instrument
How much of the symptom is present	Global impact of disease and treatment on well-being and daily function — function plus bother

Concept — *A symptom score counts what is wrong; an HRQoL instrument asks how much it matters — and only the patient can answer either reliably.*

METHODS OF ASSESSING PATIENT-REPORTED OUTCOMES

- **Questionnaires** (surveys, instruments) are preferred to **researcher-administered interviews**, which are time- and labour-intensive; instruments are cheap and practical but need rigorous development.

- **Development sequence:** define the PROs → generate and assemble items (physicians, patients, epidemiologists, psychologists; literature review for existing validated tools) → **pilot** for readability and redundancy → **psychometric analysis**.
- **Response formats:** **Likert** (categorical options — used by most PROs, easier to answer and capture) versus **VAS** (continuum).
- **Item rules:** **below an eighth-grade reading level**, no abbreviations or complex terms, **one question per item**.
- **Breadth (domains) versus depth (items per domain):** balance for patient burden, favouring breadth when the impact of disease or treatment is uncertain.

PSYCHOMETRIC PROPERTIES

Property	Meaning	Statistic and threshold
Reliability — internal consistency	Precision; agreement among items measuring one concept	Cronbach's α >0.7 good; >0.9 → redundancy
Reliability — test-retest	Reproducibility over time; retest within weeks — short enough for stability, long enough to feel fresh	Correlation >0.7
Validity — face (content)	Experts' subjective judgement that needed items are present, irrelevant absent	—
Validity — criterion	Correlation with a gold standard (e.g., new outlet-obstruction symptom tool against uroflowmetry)	$r \geq 0.7$
Validity — construct	Retrospective "gestalt" of performance, takes years; convergent (same concept, similar results) and divergent (unrelated concepts, opposite results)	Correlation coefficients
Responsiveness	Detects clinically meaningful change; often poorly assessed	Minimal clinically important difference: AUA Symptom Index 3.1 points ; otherwise $\geq \frac{1}{2}$ standard deviation

- **Validity is population-specific:** an incontinence tool validated in neurogenic bladder must be revalidated before use in CaP.

Patient-Reported Outcomes Measurement Information System

- **PROMIS**, from the NIH: standardised, flexible PRO measures; common **T-score** metric from **item response theory**; rigorous psychometrics with patient and clinician input.
- **Computer adaptive tests (CATs):** questions tailored to prior answers → **lower burden, higher precision**; fixed-length short forms where computers are limited.
- **Core domains:** fatigue, pain intensity and interference, physical function, sleep disturbance, anxiety, depression, social participation; added domains include **sexual function**, urinary incontinence and psychosocial health. The Sexual Function and Satisfaction measure is sensitive to change in men with CaP; PROMIS is reliable and responsive for stone pain, sleep disturbance in overactive bladder and bladder cancer quality of life.
- **Advantages:** **free, non-proprietary**; customisable; point-of-care administration with instant scoring; multilingual (cross-cultural comparison); adult and paediatric norms → growing use, e.g., the **CISTO study** (Comparison of Intravesical Therapy and Surgery as Treatment Options for bladder cancer).

Specific Symptom Scales

Lower Urinary Tract Symptoms

- Lower urinary tract symptom (LUTS) indices were developed for benign prostatic hyperplasia (BPH) in men but are **valid in both sexes**.
- **International Prostate Symptom Score (I-PSS)** — the best LUTS scale; seven AUA Symptom Index items (I-PSS adds a single quality-of-life item); measures severity well but **not impact on quality of life** → paired with the **BPH Impact Index**, which correlates with SF-36 general and mental health indices. Both used in large randomised trials.
- Choice of scale follows the research question.

TABLE 23.5 SELECTED PATIENT-REPORTED OUTCOMES TOOLS FOR USE PRIMARILY IN MALES WITH LUTS

Instrument	Items	Distinguishing feature
I-PSS (AUA symptom score)	7	Scored 0–35 ; gold standard PRO in BPH
BPH Impact Index (BII)	4	Impact on quality of life
International Continence Society (ICS) male short form	11	Separate voiding and continence scores
Danish Prostatic Symptom Score (DAN-PSS-I)	12	Weighted by function and bother
International Consultation on Incontinence Questionnaire (ICIQ) Nocturia Quality of Life	12 + 1	Both sexes; two themes; extra item on nocturia bother

Urinary Incontinence

- Office and urodynamic documentation is often infeasible and **artificial**: failure to demonstrate leakage in clinic does not mean the patient is dry at home or work.
- **Pad weight test**: more quantifiable during usual activities, but confounded by how pads are used and leakage around them; optimal duration unclear. **1-hour and 48-hour tests do not correlate**; **longer tests are more reproducible**. Cannot **distinguish stress from urgency** or capture **bother** → PROs are required.

TABLE 23.6 SELECTED PATIENT-REPORTED OUTCOMES TOOLS FOR USE PRIMARILY IN FEMALES WITH URINARY INCONTINENCE

Instrument	Items	Distinguishing feature
Symptoms + function + bother (disease-specific HRQoL)		
Bristol Female LUTS (BFLUTS)	33	Modified from ICS male; incontinence, voiding, storage, sexual function, quality of life; little used in men
ICIQ-FLUTS	12 (+7)	Reduced BFLUTS; extra items: 2 sexual, 5 quality of life
Incontinence Impact Questionnaire + Urogenital Distress Inventory (IIQ, UDI)	53	Used together; female-designed; short forms IIQ-7 and UDI-6 (urgency, frequency, incontinence) also perform in older men
Urge-IIQ + Urge-UDI	42	Urgency incontinence and overactive bladder (OAB) ; travel, feelings, activity, relationships, sex
OAB-q	32	Symptom-bother scale + HRQoL items; subscales 0–100, 100 = better
ICIQ	4	Frequency, amount, interference (scored) + activity item
Symptom Severity / Symptom Impact Index	16	Stress incontinence: 13 severity + 3 bother items

Predominantly HRQoL

Instrument	Items	Distinguishing feature
King's Health Questionnaire	21	10 domains ; general health, role, social, emotional, sleep; correlates with SF-36; many trials
CONTILIFE	28	Five languages; 0–100, 100 = poorer quality of life
Impact only — no symptoms or function		
Incontinence Quality of Life (I-QOL)	22	Avoidance, social embarrassment, psychosocial impact; both sexes, cross-cultural → pair with pad test, diary or symptom scale

- **What it tests:** instruments sort by what they capture — symptoms, bother, or impact alone. **Trap:** score direction differs (OAB-q 100 = better, CONTILIFE 100 = worse); I-QOL measures no symptoms.

Sexual Dysfunction

- **Difficulties:** sex differences preclude common end points; function often judged **with a partner** (problem without a regular partner or with chosen inactivity); embarrassment limits honest reporting; function is **multidimensional** — libido, arousal, erection, ejaculation/orgasm.
- **"Objective" tests** (nocturnal penile tumescence, duplex Doppler ultrasonography) occur in clinical settings unlike home and miss **psychogenic** dysfunction → **PROs are crucial**.
- **Men:** more than 20 validated instruments, mostly partner-dependent. **Erectile Dysfunction Inventory of Treatment Satisfaction (EDITS)** needs no partner but **cannot be used before treatment or without treatment**. No perfect tool — choose by setting and question.
- **International Index of Erectile Function (IIEF)** — **gold standard**, 15 items; its five erection items form the **Sexual Health Inventory for Men (SHIM)**, which **omits bother** and so is not an HRQoL measure.

TABLE 23.7 SELECTED PATIENT-REPORTED OUTCOMES TOOLS FOR USE IN MALES WITH SEXUAL DYSFUNCTION

Instrument	Items	Distinguishing feature
IIEF	15	Gold standard ; erection, libido, orgasm domains
SHIM	5	The five IIEF erection items
QOL-MED	18	HRQoL impact of erectile dysfunction (ED); assumes partner and heterosexuality
Psychological Impact of ED (PIED)	16	Sexual and emotional impact; no function
Index of Premature Ejaculation (IPE)	10	Control, satisfaction, distress
Sexual Quality of Life for Men (SQOL-M)	11	Ejaculation and ED, not libido ; correlates with IIEF satisfaction

TABLE 23.8 SELECTED PATIENT-REPORTED OUTCOMES TOOLS FOR USE IN FEMALES WITH SEXUAL DYSFUNCTION (FSD)

Instrument	Items	Distinguishing feature
Brief Index of Sexual Function for Women (BISF-W)	22	Interest/desire, activity, satisfaction; composite score added later for trials
Female Sexual Function Index (FSFI)	19	Six domains — lubrication, arousal, desire, pain, orgasm, satisfaction; summary score; most widely accepted
Derogatis Interview for Sexual Functioning (DISF)	25	Interview + self-report (15–20 min each); five domains; research, not clinic

- Far **fewer tools for female** than male sexual dysfunction; the DISF is unnecessary for most simple studies.

Health-Related Quality of Life

- HRQoL: the parts of life affected by health status — broad, multidimensional, hard to define.
Descriptions: the patient's appraisal of current functioning against the possible or ideal; the functional, psychological, social and economic impact of care. **Calman: the gap between expectations and experience.**
- **Components:** health perceptions, function, preferences, satisfaction with care; broader well-being also depends on food, shelter, response to illness and work.
- Must include **objective function and bother.**

General (generic) HRQoL	Disease-specific HRQoL
Domains common to all patients (functional and emotional well-being, overall health)	Domains relevant to one disease — e.g., invasive bladder cancer: body image, sexual and urinary function

TABLE 23.9 SELECTED HEALTH-RELATED QUALITY-OF-LIFE INSTRUMENTS THAT HAVE BEEN USED IN UROLOGIC DISEASES

Group	Instruments (items)
General	Medical Outcomes Study SF-36 (36) , SF-12 (12); Nottingham Health Profile (38); Quality of Well-being Scale (24); Sickness Impact Profile (136); EuroQol EQ-5D (5 + VAS)
Cancer-specific	Functional Assessment of Cancer Therapy-General, FACT-G (28); EORTC QLQ-C30 (30) ; Functional Living Index-Cancer (22); Cancer Rehabilitation Evaluation System-Short Form (59)
Prostate cancer	FACT-P (47); University of California, Los Angeles (UCLA) Prostate Cancer Index (20) ; Prostate Cancer Specific Quality of Life Instrument (10); Prostate Cancer Treatment Outcome Questionnaire (41); Expanded Prostate Cancer Index Composite, EPIC (36) ; Patient-Oriented Prostate Utility Scales (10)
Bladder cancer	FACT-Vanderbilt Cystectomy Index (17); Bladder Cancer Index (34) ; FACT-BL (40); EORTC muscle-invasive BLM-30 (30); EORTC superficial BLM-24 (24)
Other urological	NIH Chronic Prostatitis Symptom Index (9) ; O'Leary-Sant Interstitial Cystitis Symptom and Problem Index (23); Wisconsin Stone QOL (28); EORTC QLQ-RCC10 (10); FACT-Kidney Symptom Index, FKSI-15 (15)

Concept — Instruments are chosen by what they measure — symptom, function, bother, or impact — and are trustworthy only once reliability, validity in that population and responsiveness have been shown.

OTHER OUTCOMES OF INTEREST IN UROLOGY

Patient Satisfaction

- Used as an outcome in urological studies; general satisfaction surveys exist, **few urology-specific.**
- **Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS)** scores are **tied to Medicare reimbursement** — hospitals gain or lose **up to 1.5%**.
- **27 items: 18 core** (communication with nurses and doctors, staff responsiveness, cleanliness and quiet, pain management, medicines, discharge information, overall rating, recommendation) + 4 screening + 3 case-mix adjustment + 2 congressionally mandated.

Health Care Costs

- Accurate costing is difficult: prices vary by country and region, data are proprietary, and costs fall on different payers (patient, insurer, employer, government).

Direct	Indirect	Intangible
Actual cost of delivering care — inpatient and outpatient services (fees, staffing, equipment), drugs. Traditionally from	Lost wages of patient and caregivers, opportunity	Monetary value of pain, suffering, anxiety

Direct	Indirect	Intangible
administrative data and hospital chargemasters (inaccurate). Time-driven activity-based costing (TDABC) : map the patient's care pathway, cost each resource and the time at each step; already used for prostate cancer care	costs; depend on age, social support, severity and duration	and societal cost; very hard to measure, rarely included

KEY POINTS

- Quality is measured by structure, process and outcome; structure and process are easier, outcome most meaningful.
- Mortality is hardest but biased: lead- and length-time bias for overall mortality, attribution bias for disease-specific mortality.
- Few proxy end points meet all four surrogate criteria, yet they are used routinely.
- Standard progression and complication systems should be used, but do not remove design bias.
- Frailty, functional status and comorbidity are confounders with standard tools.
- Use validated, reliable patient-centred instruments whenever possible.